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An Efficient Deep Learning Approach for Jute Pest Classification Using Transfer Learning

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***Abstract*— Jute is considered as one of the most vital crops in the world. For some countries jute is the principal source of earnings and GDP. One of the primary elements influencing jute yield is jute pests. Accurate pest identification makes it possible to take prompt preventative action to minimize financial losses. Considering the fact, to classify jute pests, the study suggests different jute pest classification models, which are based on transfer learning. The best model offers high performance and resilience. A UCI-validated dataset comprising 7235 images has been utilized in the analysis. The dataset encompasses images classified into 17 distinct jute pest classes. The dataset is already divided into three categories train, test and validation. To increase the dataset size, data augmentation is applied to the training set. To improve performance, all the models were integrated with the transfer learning model. VGG16, ResNet101, DenseNet201, InceptionV3, Xception, and MobileNetV2 were used to train the parameters on the publicly available ImageNet dataset followed by some customized dense layers. The models were assessed using different types of metrics, including confusion matrix, F1 score, precision, and recall. Compared to other models DenseNet201 outclassed other models, acquiring 97% accuracy. The fundamental information and technical support for jute pest classification are provided by this study.**

Keywords*— *Jute Pest, Data Augmentation, DenseNet201, Transfer Learning.

I. Introduction

One of the main global food crops for human consumption, jute is essential to global food security. Jute, one of the main cash crops in Bangladesh, is also environmentally beneficial. In Bangladesh's industrial sector of the economy, jute plays a significant role [1]. When it comes to the jute industry, Bangladesh has continuously held an advantage over China and India [2]. Nearly 85% of the world's jute production and cultivated land is contributed by Bangladesh and India, the two main jute-growing nations. Bangladesh accounts for over 39% of the world's entire supply of raw jute, grown on 39% of the total global jute land, with an average production of 15.32 q/ha between 1961 and 2002 [3]. The jute has some notable characteristics. The jute kind has a unique capacity to eliminate heavy metals from the atmosphere, including lead, copper, cadmium, zinc, and mercury. Moreover, jute is less expensive, more reusable, resilient, and adaptable than synthetic fibers [4]. Jute has several other important benefits, like being ago-based, produced annually, renewable, and biodegradable [5]. Jutes produce oxygen to purify the air, boost soil fertility, and protect the ozone layer by absorbing CO2 [6]. Similar to many other crops, jute is susceptible to various pests. Around 40 kinds of insects and mites are thought to affect jute [7]. These pests pose a major obstacle to the production of this adaptable crop of fiber. These pests have the potential to seriously harm jute plants, compromising the yield's quality and quantity.

Jute pests have an effect outside of the fields as well, affecting the jute industry's overall economic sustainability. To reduce these issues and maintain a strong jute industry, effective pest management and detection techniques are essential. A variety of control methods are frequently used by farmers to protect jute crops from damage caused by pests. By effectively classifying pests, this study offers a useful solution that helps farmers apply pesticides [27] that are specifically designed to combat the pests that are posing a threat to their jute crops.

An important part of the agricultural economy is jute, a cash crop. However, pest infestations are a serious threat to the production of jute, resulting in lower yields and financial difficulties for farmers. Our research attempts to offer an effective solution for early pest detection and classification, eventually increasing agricultural productivity, by utilizing deep learning techniques and transfer learning. Our research aims to promote environmentally friendly pest control methods and lessen reliance on chemical pesticides by utilizing cutting-edge deep learning approaches. By helping farmers implement targeted interventions, an effective pest classification system can help create a more sustainable and environmentally friendly agricultural ecosystem.

II. Related Work

Our world is changing quickly, and artificial intelligence (AI) is having an impact on almost every industry. AI is streamlining processes, increasing productivity, and spurring innovation in everything from self-driving cars to medical diagnosis. Machine learning and deep learning model development is one of the main forces behind the growth of AI. Without explicit programming, these models are able to learn from data and generate predictions. This makes them perfect for tasks like pattern recognition in large datasets or object identification in images that are too complicated or time-consuming for humans to perform.

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Machine learning allows computers to learn without being explicitly programmed. Just like humans learn from experience, machine learning algorithms can identify patterns and make predictions from data. This makes them incredibly versatile, and they're used in everything from facial recognition software or rice pest classification [8] to the detection of plant disease. Shape features and machine learning techniques like artificial neural networks (ANN), support vector machines (SVM) [9]-[12], k-nearest neighbor (KNN) [13] and then again SVM [18] models were used in additional research to produce the best detection performance with the least amount of computation time. On the other hand, deep learning methods take into account a taxonomy of various real world tasks, encompassing both supervised and unsupervised learning paradigms [26]. In the short time transfer learning has taken to become a widely used technique in the field of deep learning, it has become increasingly popular. It is used as feature extractor [30]. It can be effectively used for works like different image classification [29] and malware classification [28] as well. A greedy deep learning method was introduced in [15] for medical disease analysis in that paper Wu et al. proposed an weighted mechanism that connect sample with the dictionary. Islam et al. [19] use transfer learning with VGG16 to detect potato disease, where they got higher accuracy than any other research had conducted before. With 80% of the training dataset and 20% of the dataset for training, they achieved an accuracy of 90.43%. Kasinathan et al. [20] conducted a research on dataset of 24 classes, Using the customized CNN model, the highest classification rates of 91.5% and 90% were attained for the nine and 24 class insects respectively. A different approach was taken by Ung et al. [21], using CNN based that includes feature pyramid, fine-grained models and attention. Sourav et al. developed a DCNN model for a dataset consisting of 4 major jute pest classes. They obtained 95.86% accuracy. However, it is clear from doing a literature review that a few studies have been done on pest classification or identification. Working with imbalance dataset [16] is also should be taken into consideration. There were multiple types of research on pest classification, like multi-class pest detection [14], these previous research had good precision, recall, and accuracy as there were less number of classes. Utilizing different deep learning models, previous studies have mostly concentrated on the identification of four different types of jute pests [22]. In [23] DCTN model with VGG16 architecture found an outstanding performance achieving 99% test accuracy. Yolov5 model was used for detecting pests by Ahmed et al. [25]. Other pre-trained models were not used in their investigations. Thus, the goal of this research is to increase accuracy by adding more jute pest classes.

III. Material and Methods

There are four steps in our proposed approach that go hand in hand. In order to train and assess the DL models, we first gather images of pests. In order to increase the number of samples in the dataset, we secondly preprocess the entire dataset using augmentation. The process of artificially expanding a training dataset by slightly modifying preexisting images in accordance with predetermined parameters is known as image data augmentation. Third, we use transfer learning to use the various pre-trained deep learning models. Lastly, we evaluate the model using various metrics in order to choose the best model. In fig. 1 the overall workflow of the proposed approach has been shown

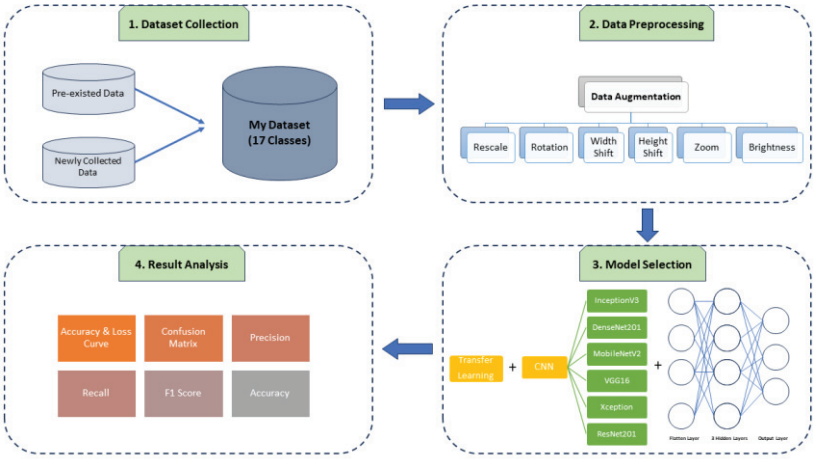


Fig. 1: Overview of Research Methodology

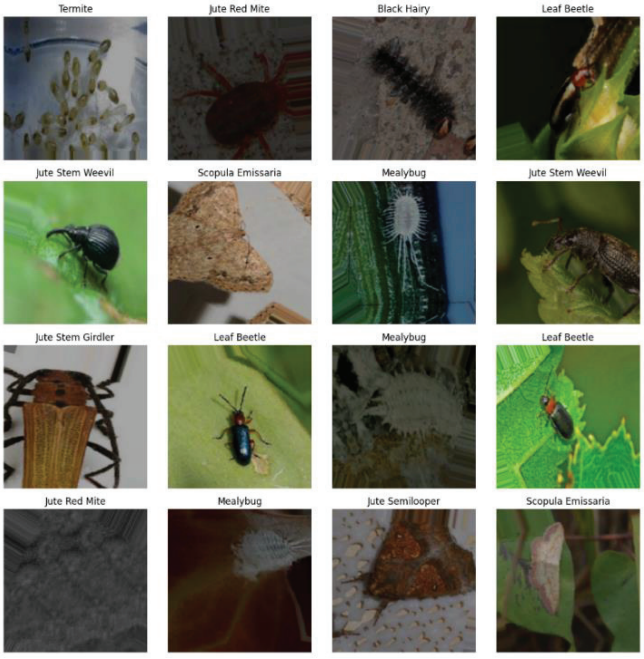


Fig. 2: Sample Image of the Training Dataset

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A. Dataset Collection

The Jute Pest Dataset [32] is validated by UCI and this website provides open source datasets. So, the dataset is available on the UCI dataset website, and it is free to download. The dataset comprises 7235 properly stored photos of different types of jute pests. The dataset has 17 classes. This varied picture collection contributes to advances in agricultural research and pest control techniques by providing a thorough resource for researching and comprehending the different pests that harm jute crops. This diverse collection of images offers a comprehensive resource for studying and understanding the various pests affecting jute crops, contributing to advancements in agricultural research and pest management strategies. All the images are distributed in three categories train, test and validation. Fig. 2 gives insight of sample training images of jute pests. Here 16 classes images are shown.

B. Data Preprocessing

Increasing the quantity of training data is crucial for improving deep learning model performance, yet gathering sufficient data is challenging. Insufficient data often leads to overfitting during training. Moreover, Data imbalance can skew model predictions and decrease accuracy. Data augmentation, particularly geometric transformations, addresses this issue by generating additional training samples. In our study, images were standardized to 224 × 224 × 3 size and converted to RGB format for consistency as previously not all the images were in RGB format. ImageDataGenerator was then utilized for augmentation, applying six techniques: rotation, height and width shift, zoom, and brightness adjustment. These transformations expose the model to variations in object orientations, perspective, scale, and lighting conditions, enhancing its generalization ability. Height and width shift techniques shift the image vertically or horizontally, simulating changes in perspective. Zoom alters the scale of the image, helping the model recognize objects at different distances and sizes. Brightness adjustment changes overall brightness, aiding the model's adaptation to varying lighting conditions.

C. Model Selection

Here we used transfer learning with six different pre-trained CNN models followed by customized dense layers and softmax layer. All these are briefly discussed in this part.

1) Transfer Learning: Transfer learning is a machine learning technique in which a model that has been trained on one task is modified to execute another related task. The term "transfer learning" describes the application [17] of skills or patterns acquired in one domain or endeavor to another related domain or issue. The core idea behind transfer learning is to leverage knowledge from related domains to enhance the effectiveness of a target task, particularly when there is limited labeled data or computational resources available. Transfer learning, commonly used in convolutional neural networks, allows models to be retrained with small datasets. By freezing the pre-trained layers, the model retains the knowledge gained from large datasets like ImageNet while adapting to new tasks. This approach, employed in our study by freezing the last layers, enables the model to serve as a feature extractor for the new task, enhancing performance and accelerating convergence when labeled data is scarce.

2) Xception: Xception's use of depth-wise separable convolutions improves the effectiveness of feature extraction and increases the expressiveness of the model. Xception is a powerful and effective network that outperforms on benchmark datasets like ImageNet for tasks like object detection, semantic segmentation, and image classification thanks to its many convolutional and depth-wise separable convolutional layers. Its effectiveness is also seen in medical imaging, where it makes illness analysis and diagnosis easier.

3) MobileNetV2: MobileNetV2 is a lightweight and efficient convolutional neural network architecture, particularly suitable for mobile and edge devices. It enhances both accuracy and speed compared to its predecessor, MobileNet, by optimizing the trade-off between performance and model size. Utilizing depthwise separable convolutions, shortcut connections, and linear bottlenecks, MobileNetV2 achieves high computational efficiency without sacrificing expressiveness, making it ideal for real-time image processing and object detection on resource-constrained devices.

4) DenseNet201: DenseNet-201 is a strong convolutional neural network architecture that uses a dense connectivity structure to address training problems in deep networks. In addition to solving vanishing gradient issues, this technique enables efficient feature reuse. DenseNet-201 is an extremely efficient deep structure for image classification tasks, collecting complicated hierarchical information with ease. It is composed of dense and transition blocks. It clearly has significant generalization capabilities, especially with sparsely labeled data. Comparative analyses show that it is competitive with state-of-the-art architectures in terms of accuracy, efficiency, and computing cost. In general, DenseNet-201 proves to be a solid and efficient option in this study.

5) Flatten Layer: The flattening process is essential because it sets up the data for the dense layers that follow when creating custom fully connected layers on top of a trained model. We can then add more layers and fine-tune the model for a new problem while utilizing the knowledge gained by the pre-trained layers. This helps to adapt the pre-trained features to the specific requirements of our task.

6) Fully Connected Layers: To enable the model to learn high-level abstractions and representations from the flattened features, these fully connected layers are present. As the number of neurons in each layer is reduced, the information is gradually distilled and condensed, preserving key characteristics and lowering the complexity of the model. Here we added custom fully connected layers on top of the flattened output from the pre-trained model. Each line corresponds to a Dense layer, specifying the number of neurons (units) and the activation function for each layer.

7) Softmax Layer: The output values are normalized into probabilities using the softmax activation function, which

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guarantees that the total of the probabilities for all classes equals 1. This makes it appropriate for situations where the model must choose between several classes. The class with the highest probability is the one that is anticipated. We added the final output layer to our custom model. The last layer with 17 neurons, corresponding to the number of classes or categories in our classification problem. The activation function used here is softmax, which is commonly employed in multi-class classification tasks.

D. Evaluation Metrics

1) Confusion Matrix: The confusion matrix categorizes model predictions into True Positives, True Negatives, False Positives, and False Negatives, enabling the calculation of key metrics like accuracy, precision, recall, specificity, and the F1 score. It offers a detailed analysis of a classification model's strengths and weaknesses, making it particularly useful when dealing with unequal class distributions.

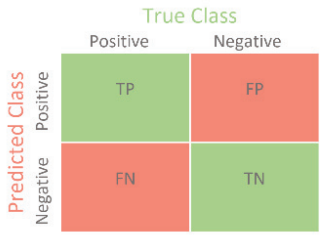


Fig. 3: Cofusion Matrix

2) Precison: Precision, which is the percentage of accurately predicted positive cases among all instances anticipated as positive, is a metric used to assess the accuracy of positive predictions. Formula is given below:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

3) Recall: The fraction of accurately predicted positive instances among all actual positive instances is known as recall, and it measures the model's capacity to catch all positive cases. Formula is given below:

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

4) F1 Score: A balanced rating that blends recall and precision is the F1 Score. In an effort to strike a balance between recall and precision, it offers a single number those accounts for both false positives and false negatives.

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (3)$$

IV. Experimental Results

The result part is discussed in two parts. First, we will discuss the performance and then we will discuss a comparative analysis with existing studies.

A. Performance

The outcomes of 6 distinct customized pre-trained models are compared and analyzed in this section of the study article. The accuracy curve, loss curves, confusion matrix, precision, recall, F1 score, and accuracy, for each of the six models are used to evaluate the models and decide which one is the best for the jute pest classification.

In Fig. 4 training and validation accuracy curve and training and validation loss curve are given in order: i) VGG16, ii) DenseNet201, iii) Xception, iv) ResNet101, v) InceptionV3 and vi) MobileNetV2.

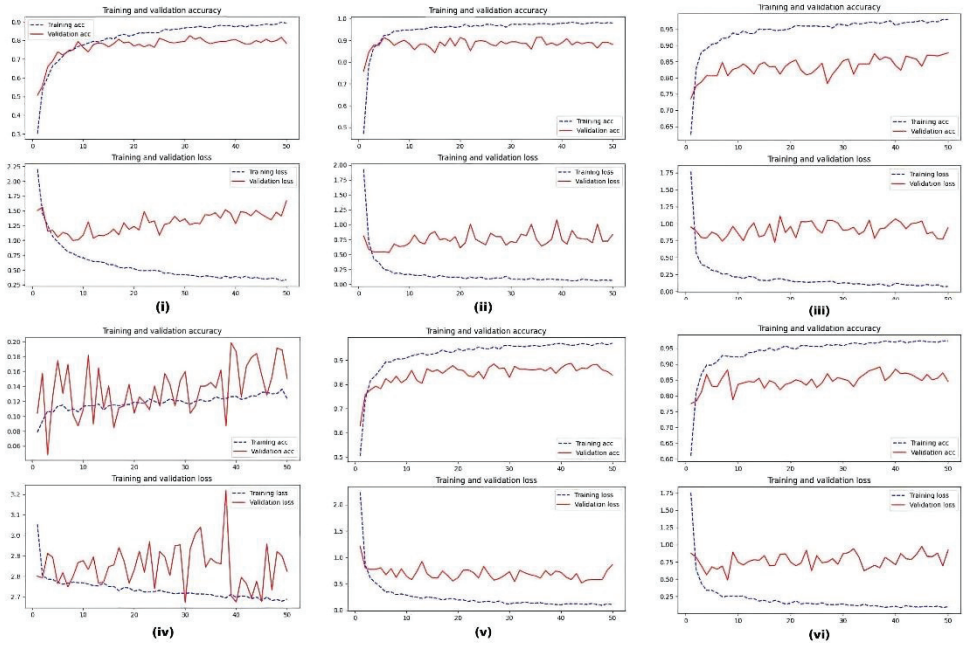


Fig. 4: Curve: Training, Validation Accuracy and Training, Validation Loss for all the Models.

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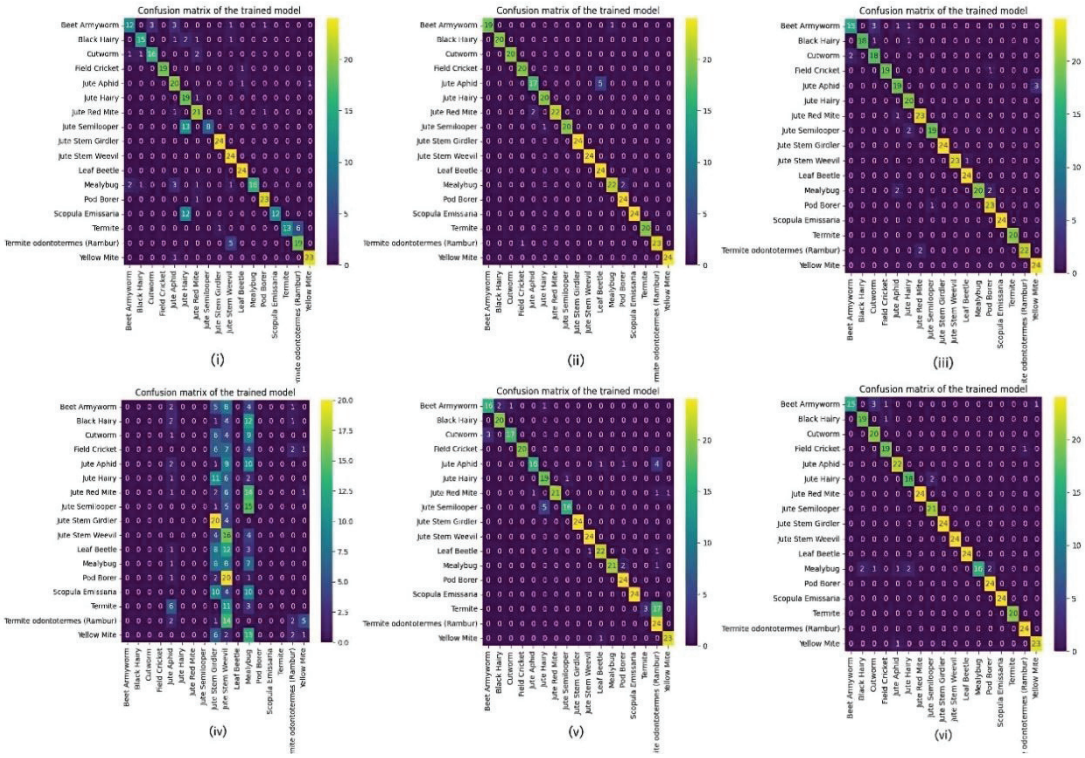


Fig. 5: Confusion Matrices for all the Models

On the other hand, Fig. 5 the confusion matrices are given in order: i) VGG16, ii) DenseNet201, iii) Xception, iv) ResNet101, v) InceptionV3, vi) MobileNetV2.

After applying all the models based on our research approach, we can see our results in Table 1. DenseNet201 has achieved the highest accuracy as well as the precision, recall and f1 score. All of them are 97%. The next good performed models are MobileNetV2 and Xception. Both models' f1 scores are 95% and 94% respectively. Other models' performances are shown in the tables.

TABLE I. MODELS' ACCURACY

Models	Accuracy	Precision	Recall	F1 Score
DenseNet201	97%	0.97	0.97	0.97
MobileNetV2	95%	0.96	0.95	0.95
Xception	94%	0.94	0.94	0.94
InceptionV3	88%	0.92	0.88	0.87
VGG16	81%	0.86	0.81	0.81
ResNet101	13%	0.05	0.13	0.06

B. *Discussion*

In this study, we compared and contrasted several image classification models that have been applied to various agricultural research papers. Every study project examined different facets of agricultural pests and diseases, using different models to produce precise classification. While our dataset had certain limitations like data imbalance, shortage of pest types etc., we achieved favorable results. For future endeavors, addressing the data imbalance is imperative to optimize outcomes further. T4 GPU was utilized for training. Training time amounted to 5 hours. For future endeavors, optimization techniques such as model pruning or quantization could be explored to enhance computational efficiency. Specific models were selected over others for their superior performance in feature extraction and transfer learning tasks, providing enhanced accuracy and efficiency tailored to the specific problem domain.

Using DenseNet201, we were able to obtain an impressive 97.00% accuracy on Jute Pest. Studies like Kasinathan et al.'s Field Crop Insects with nine and twenty-four classes and Sourav et al.'s Jute Pests with four classes demonstrated the trade-off between specificity and comprehensiveness due to the variation in the number of classes. There were notable performance differences, with Jiang et al.'s Rice Pests study utilizing VGG16 to achieve an accuracy of 88.68%, indicating possible difficulties in specific crop situations. The field's dynamic nature is highlighted by the way models evolve over time, as demonstrated by Li et al.'s use of YOLO-JD in 2022. Furthermore, the models' transferability, demonstrated by the way that VGG16 is used in various studies, highlights the ways in which specific architectures may be able to adapt to a range of agricultural challenges.

TABLE II. COMPARATIVE ANALYSIS WITH EXISTING WORKS

Author & Year	Types	Number of Classes	Model	Accuracy
This Work	Jute Pest	17	DenseNet201	97.00%
Li et al. (2022) [24]	Jute Diseases and Pests	17	YOLO-JD	96.63%
Talukder et al. (2023) [31]	Jute Pest	17	DenseNet201	99.00%
Sourav et. al (2022) [22]	Jute Pests	4	VGG19	95.86%

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In [31], Talukdar et al. achieved superior results with DenseNet201 on a dataset with a similar number of classes to ours. However, while both datasets share similarities in the number of classes, they differ slightly in terms of image count and pest types. All the studies listed in the comparison table utilized different datasets, despite having a similar count of classes. Importantly, there has been no prior research conducted using the dataset employed in this study.

V. Conclusion and Future Directions

In conclusion, we found that our suggested deep learning strategy, enhanced by transfer learning, was very successful in precisely and quickly classifying jute pests. The model's consistent ability to identify pests was demonstrated when its accuracy exceeded the baseline. Transfer learning significantly decreased computational load and expedited training. This flexible and modular model has a great deal of potential for practical use, which could result in early pest detection, better pest management techniques, and eventually higher yields and financial gains for jute growers. To further improve and validate the model's impact on Jute pest management and transform Jute farming practices, more research on pest diversity, environmental factors, and mobile app development is needed.

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